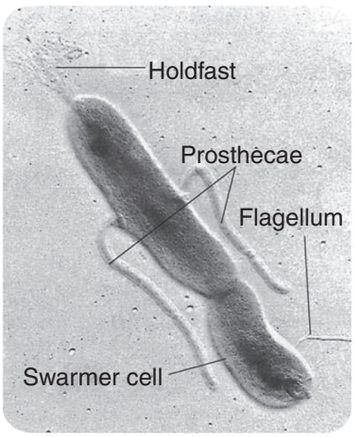


Figure 7.9: Prosthecate bacteria

Function:

- Prosthecae increase surface area for absorption of nutrients from the dilute aquatic environment.
- Helps in adhesion.
- Some prosthecae develop bud at the tip and helps in asexual reproduction.

Stalk

It is a nonliving ribbon like tubular structure. It is formed by excretory product of bacteria. Some of the stalked bacteria are *Gallionella, Planctomyces* (Figure 7.10).

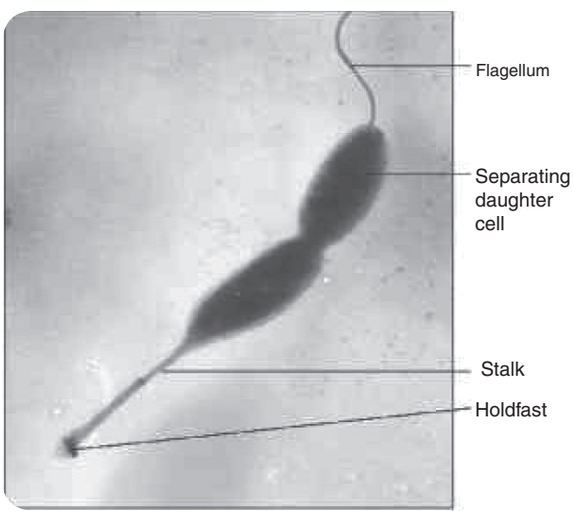


Figure 7.10: Stalked bacteria

Function:

- Stalk helps in attachment of cells to solid surface.

7.3 Cell Envelope of Bacteria

Cell envelope is an external covering that lies outside the cytoplasm. It is composed of two or three basic layers: the cell wall, the cell membrane and in some bacteria the outer membrane.

7.3.1 Structure of Prokaryotic Cell Wall

Prokaryotic cells almost always are bounded by a chemically complex cell wall. Cell wall lies beneath the external structures (capsules, sheaths and flagella). Cell wall lies external to the plasma membrane (cell membrane). Cell wall of eubacteria is made up of **peptidoglycan** or **murein**, whereas that of Archaeobacteria is composed of proteins, glycoproteins or polysaccharides. A few genera such as *Methanobacterium*, have cell walls composed of **pseudomurein**, a polymer whose structure superficially resemble eubacteria peptidoglycan of eubacteria but differs markedly in chemical composition.(Note: Ordinary or typical bacteria are sometimes called eubacteria to distinguish them from the phylogenetically distinct group known as archaeobacteria). Peptidoglycan is a cross linked polymer of enormous strength and rigidity. It is a polymer composed of many identical subunits (Figure 7.11). Peptidoglycan differs somewhat in composition and structure from one species to another, but it is basically a polymer of N-acetylglucosamine(NAG), N-acetylmuramic acid(NAM), L-alanine, D-alanine, D-glutamate, and a diamino acid (LL- or meso-diaminopimelic acid, L-lysine, L-ornithine, or L- diaminobutyric acid).

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